

One-step synthesis of N-doped hierarchical porous carbon from rice husks via urea-mediated dual activation for high-performance aqueous zinc-ion capacitors

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Highlights

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N-doped porous carbon (NKC) from rice husks via one-step pyrolysis with K₂CO₃/urea.

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Urea serves as both nitrogen dopant and mesopore-forming agent.

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NKC based ZIC achieves 354.3 F g⁻¹ at 0.5 A g⁻¹ and 57.9 % retention at 20 A g⁻¹.

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Tailored porosity and N-doping enable superior kinetics and performance of ZIC.

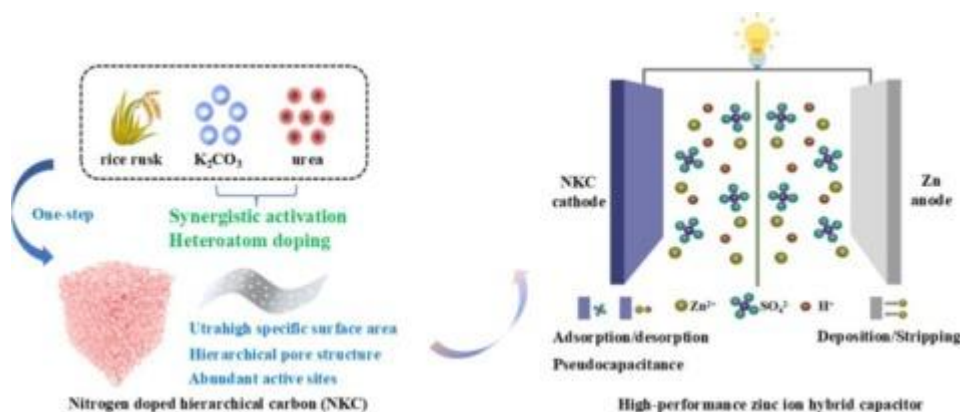
Abstract

The development of high-performance aqueous zinc-ion capacitors (ZICs) is hindered by the capacity-rate trade-off in conventional carbon cathodes due to unsuitable pore structures. Herein, we report a facile one-step synthesis of N-doped hierarchical porous carbon (NKC) using rice husks (RHs), potassium carbonate (K₂CO₃) and urea. Crucially, K₂CO₃ primarily activates micropores of RH-derived carbon, while urea serves as both a nitrogen dopant and a pore-forming agent, generating abundant mesopores on the carbon sheets during pyrolysis. The synergistic pore engineering yields NKC with an ultrahigh specific surface area (3016.8 m² g⁻¹), nearly double that of undoped KC sample. The tailored multiscale porosity can offer micropores for charge storage and mesopores/macropores for rapid ion transport. N-doping further enhances the

conductivity, wettability and active sites, and reduces the adsorption energy of Zn^{2+} on the oxygen functionalities. Benefiting from the synergistic effect, the optimized NKC based ZIC presents a high specific capacitance of $354.3 \text{ F g}^{-1}/155.8 \text{ mAh g}^{-1}$ at 0.5 A g^{-1} , excellent rate performance (57.9% retention at 20 A g^{-1}), and distinguished energy/power density (154.5 Wh kg^{-1} at 495.8 W kg^{-1}), surpassing most reported carbon-based ZICs.

Electrochemical characterizations, ex-situ structural analysis and theoretical calculations collectively elucidate the enhanced kinetics and storage mechanism, confirming the critical role of the synergistic pore and heteroatom engineering. This study demonstrates an efficient approach to fabricate high-performance biomass-derived carbon cathode for practical ZICs.

Graphical abstract



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Introduction

The rapid growth of modern technologies including portable electronics, electric vehicles and smart grid energy storage has driven substantial progress in developing economical and high-performance electrochemical energy storage system [1,2]. As innovative hybrid energy storage devices, metal-ion capacitors synergistically integrate the advantageous features of two distinct mechanisms, leveraging faradaic reactions from battery-type anodes to achieve exceptional energy storage capabilities while maintaining rapid charge-discharge characteristics through surface ions adsorption/desorption process of capacitive cathode [3,4]. Among these, aqueous zinc-ion capacitors (ZICs) have emerged as a particularly promising candidate, attributed to the superior characteristics of metallic zinc anode including its favorable working potential (-0.76 V), remarkable specific capacity (820 mAh g^{-1}), and reliable electrochemical reversibility in aqueous electrolytes, along with inherent operational safety [5,6]. However, the performance of ZICs is often constrained by

the capacitor-type cathode, which limits the total specific capacity. Consequently, significant efforts have been dedicated to optimizing cathode materials to enhance the overall device performance [[7], [8], [9], [10], [11], [12]]. Carbon-based materials, comprising activated carbon, graphene, and carbon nanotubes, demonstrate considerable potential as cathodes for ZICs because of their large surface areas and/or distinguished electrical conductivity [[13], [14], [15]]. Nevertheless, their unsuitable pore architecture with excessive micropores or macropores typically results in a trade-off between capacity and rate performance.

To address this limitation, hierarchically porous carbon (HPC) integrating micropores, mesopores and macropores have attracted significant attention due to their synergistic merits [16]. Micropores primarily contribute to large specific surface area for charge storage, while mesopores act as efficient ions transport channels, and macropores serve as electrolyte reservoirs, collectively facilitating rapid ion transport kinetics during electrochemical processes [17,18]. Thus, multiscale pore engineering proves essential for achieving concurrent enhancements in both specific capacitance and rate capability [[19], [20], [21]]. Biomass, particularly rice husks (RHs), offers a promising, low-cost carbon source for HPC synthesis [22,23]. RHs represent a major agricultural waste from rice milling, with global annual production exceeding 100 million tons [24]. Critically, their cellular structure contains approximately 20 wt% of non-crystalline silica nanoparticles distributed within intercellular regions and cell wall matrices, which can serve as natural templates for synthesizing carbon materials with multiscale porosity [25]. Despite these advantages, the electrochemical performance of RH-derived carbons, especially when employed as cathodes in ZICs, often remains suboptimal [22,26,27]. The limitation primarily stems from inadequate control over the pore architecture, often resulting in a mismatch for transport and storage of hydrated Zn^{2+} , and a lack of sufficient electrochemically active sites.

Consequently, simultaneous engineering of the pore hierarchy and surface chemistry is highly desirable to overcome the above issues, yet it remains a significant challenge through a simple process. While heteroatom doping, particularly nitrogen (N) doping, is a well-established strategy to enhance the electrochemical activity of carbon materials by improving conductivity, wettability, and introducing pseudocapacitive sites [[28], [29], [30], [31]]. Besides, the pyrolysis of nitrogen precursors may concurrently induce pore formation through gas evolution, which would enable the synchronous optimization of surface chemistry and hierarchical porosity [32]. However, such a synergistic and simplified approach, tailored for enhancing the performance of RH-derived carbon in ZICs, has been scarcely explored.

Herein, we report a facile and effective one-step synergistic strategy for in-situ preparation of nitrogen-doped hierarchical porous carbon (N-doped HPC, denoted as NKC) using RHs as carbon resources, potassium carbonate (K_2CO_3) as activation agents and urea as nitrogen resources. The approach integrates carbonization, activation, and doping into a single thermal process. The innovation of this work lies in the deliberate and synergistic combination of K_2CO_3 and urea to achieve complementary pore-structuring and functionality. We systematically demonstrate that while K_2CO_3 primarily creates microporosity, the incorporation of urea plays a crucial dual role as both an efficient nitrogen source for doping and in-situ mesopore-forming agent via gas release during pyrolysis. The synergistic effect yields a material (NKC) with an ultrahigh specific surface area ($3016.8 \text{ m}^2 \text{ g}^{-1}$) and a well-defined hierarchical pore network. The ZIC assembled with NKC cathode delivers an ultrahigh specific capacitance (354.3 F g^{-1} at 0.5 A g^{-1}) and distinguished rate capability (57.9 % retention, $0.5\text{--}20 \text{ A g}^{-1}$), outperforming most previously-reported carbon cathode based ZICs. A comprehensive investigation combining electrochemical analysis, ex-situ characterization, and DFT calculations was conducted to elucidate the underlying enhancement mechanisms, confirming the critical importance of our synergistic pore and heteroatom engineering strategy. This work provides an efficient and scalable paradigm for transforming agricultural waste into high-performance electrode materials for advanced energy storage devices.

Materials

The RHs were achieved from Hongqiling Farm (State Farm of Heilongjiang Agricultural Reclamation). K_2CO_3 (AR), urea (AR), sodium hydroxide (NaOH, AR) and sulfuric acid (H_2SO_4 , 98 %) was gained by Sinopharm Reagent Co., Ltd. (China). Zinc foil (20 μm) was bought from Beijing Zhongnuo New Materials Technology Co., Ltd. (China). Zinc sulfate heptahydrate ($\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$, AR) was acquired from Tianjin Fuchen Chemical Regent Co., Ltd. (China). The hydrophilic PTFE filter paper with pore size of 0.45 μm

Results and discussion

The preparation procedures are schematically illustrated in Fig. 1a. The RHs primarily composed of lignin and cellulose were utilized as carbon resources, and homogeneously mixed with K_2CO_3 as activation agents and urea as nitrogen resources, followed by thermal treatment at 800°C . The one-step strategy simultaneously accomplishes carbonization, chemical activation and heteroatom doping for advanced material fabrication. Specifically, during pyrolysis, the inherent silica ($\sim 20 \text{ wt\%}$) within the

Conclusion

In summary, a facile one-step strategy was developed to synthesize nitrogen doped hierarchical porous carbon (NKC) from abundant rice husk biomass, demonstrating its exceptional performance as a cathode for aqueous zinc-ion capacitors (ZICs). By synergistically combining K_2CO_3 activation with urea-mediated dual-function engineering (simultaneous N-doping and in-situ pore generation), the resulting NKc features a well-defined hierarchical architecture comprising micropores, mesopores, and

CRediT authorship contribution statement

Qinqin Zhou: Writing – original draft, Validation, Methodology, Funding acquisition, Conceptualization. **Lixing He:** Visualization, Methodology, Investigation. **Chuanjin Wang:** Software. **Ling Li:** Investigation. **Qingyan Bai:** Investigation. **Xiaoshuai Ma:** Validation. **Siyu Hao:** Validation. **Peng Hu:** Funding acquisition. **Jinshu Wang:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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